Program 3: Working with Selenium: Maven Project To Connect To Web Servers Using Selenium Dependency

1.Sauce.demo webpage

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -Darchet
ypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenSeleniumApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.217 s
[INFO] Finished at: 2026-03-19T14:17:33+05:30
[INFO] ------------------------------------------------------------------------
sumanth@sumanth-VirtualBox:~/devops$
```

Step 2: Before we connect to web servers we need to have chrome driver installed. To download chrome driver

To check the chrome version type the command **google-chrome –version** in the terminal

extract the version and store it in a variable CHROME_VERSION

extracts and transorms JSON data

LATEST_DRIVER=$(curl -s "https://googlechromelabs.github.io/chrome-for-testing/latest-patch-versions-per-build.json" | jq -r ".builds[\"$CHROME_VERSION\"].version")

do install curl using the command

sudo apt install curl

wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip

Extract and move it to /usr/local/bin/:

unzip chromedriver-linux64.zip

sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver

sudo chmod +x /usr/local/bin/chromedriver

Check if ChromeDriver is installed correctly:

```
sumanth@sumanth-VirtualBox:~/devops$ chromedriver --version
ChromeDriver 145.0.7632.117 (de4e5d6e13b995f5e04137970c0e8ec456017467-refs/branch-heads/7632@{#3101})
sumanth@sumanth-VirtualBox:~/devops$
```

Step 3: edit the pom.xml file and add the dependencies
The pom.xml has the code

Open *pom.xml ~/Devops/MyMavenSeleniumApp01 Save

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenSeleniumApp01</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenSeleniumApp01</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <!-- Source: https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.41.0</version>
    <scope>compile</scope>
</dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
```

```
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>


        </plugin>


</plugins>
</build>
</project>
```

Step 4: **<u>Edit the App.java</u>**

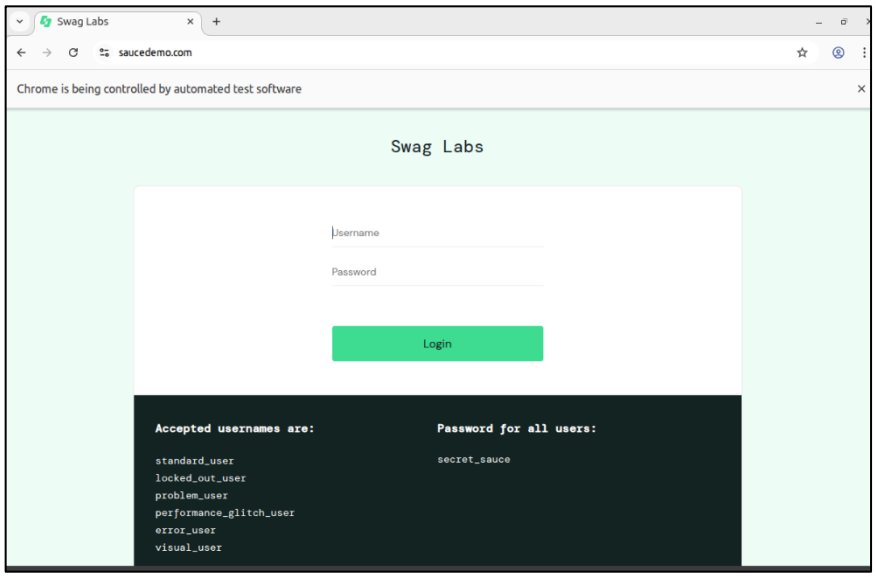
```
Open    *App.java    ~/Devops/MyMavenSeleniumApp01/src/main/java/com/example    Save
1 package com.example;
2 import org.openqa.selenium.By;
3 import org.openqa.selenium.WebDriver;
4 import org.openqa.selenium.chrome.ChromeDriver;
5 public class App
6 {
7     public static void main( String[] args )
8     {
9         WebDriver driver=new ChromeDriver();
10        driver.get("https://www.saucedemo.com/");
11        driver.manage().window().maximize();
12        driver.findElement(By.id("user-name")).sendKeys("standard_user");
13        driver.findElement(By.id("password")).sendKeys("secret_sauce");
14        driver.findElement(By.id("login-button")).click();
15    }
16 }
```

Step 5: build and run the maven application

Type **mvn clean install** in the terminal

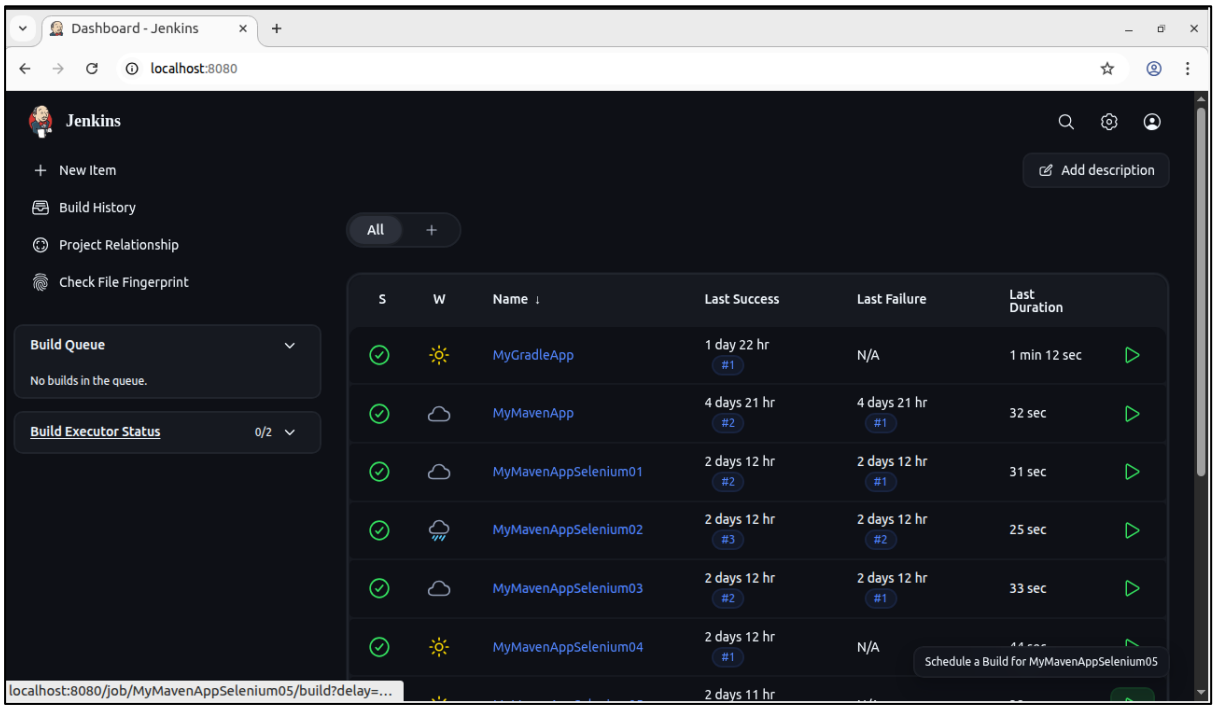
```
sumanth@sumanth-VirtualBox: ~/devops/MyMavenSeleniumApp01
sumanth@sumanth-VirtualBox:~/devops/MyMavenSeleniumApp01$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenSeleniumApp01 ---
[INFO] Deleting /home/sumanth/devops/MyMavenSeleniumApp01/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/devops/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/devops/MyMavenSeleniumApp01/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/devops/MyMavenSeleniumApp01/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenSeleniumApp01 ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/sumanth/devops/MyMavenSeleniumApp01/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenSeleniumApp01 ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.01 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.0.1:jar (default-jar) @ MyMavenSeleniumApp01 ---
[INFO] Building jar: /home/sumanth/devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenSeleniumApp01 ---
[INFO] Installing /home/sumanth/devops/MyMavenSeleniumApp01/target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar to /home/sumanth/.m2/repository/com/example/MyMavenSeleniumApp01/1.0-SNAPSHOT/MyMavenSeleniumA
pp01-1.0-SNAPSHOT.jar
[INFO] Installing /home/sumanth/devops/MyMavenSeleniumApp01/pom.xml to /home/sumanth/.m2/repository/com/example/MyMavenSeleniumApp01/1.0-SNAPSHOT/MyMavenSeleniumApp01-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.900 s
[INFO] Finished at: 2026-03-19T14:22:18+05:30
[INFO] ------------------------------------------------------------------------
sumanth@sumanth-VirtualBox:~/devops/MyMavenSeleniumApp01$
```

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"** it opens **saucedemo.com** and we can login using the credentials



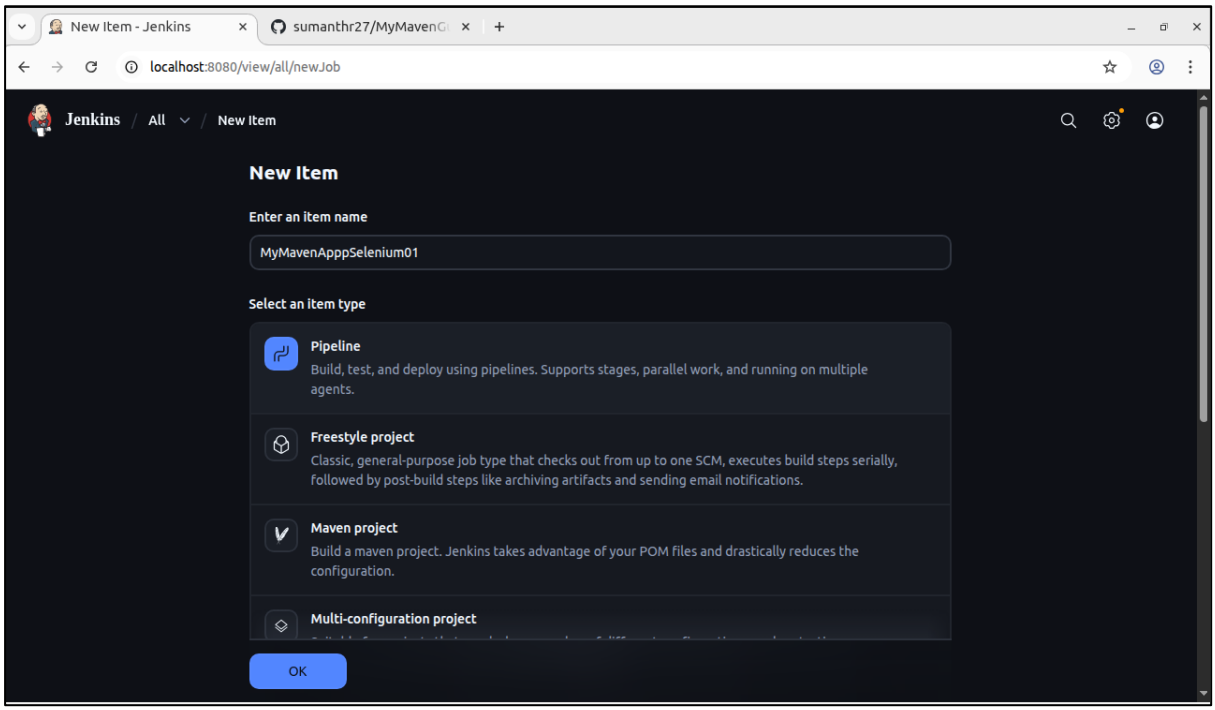
Jenkins Pipeline for MyMavenSelenium01

Step 1: Open the Jenkins Dashboard in order to create a new item

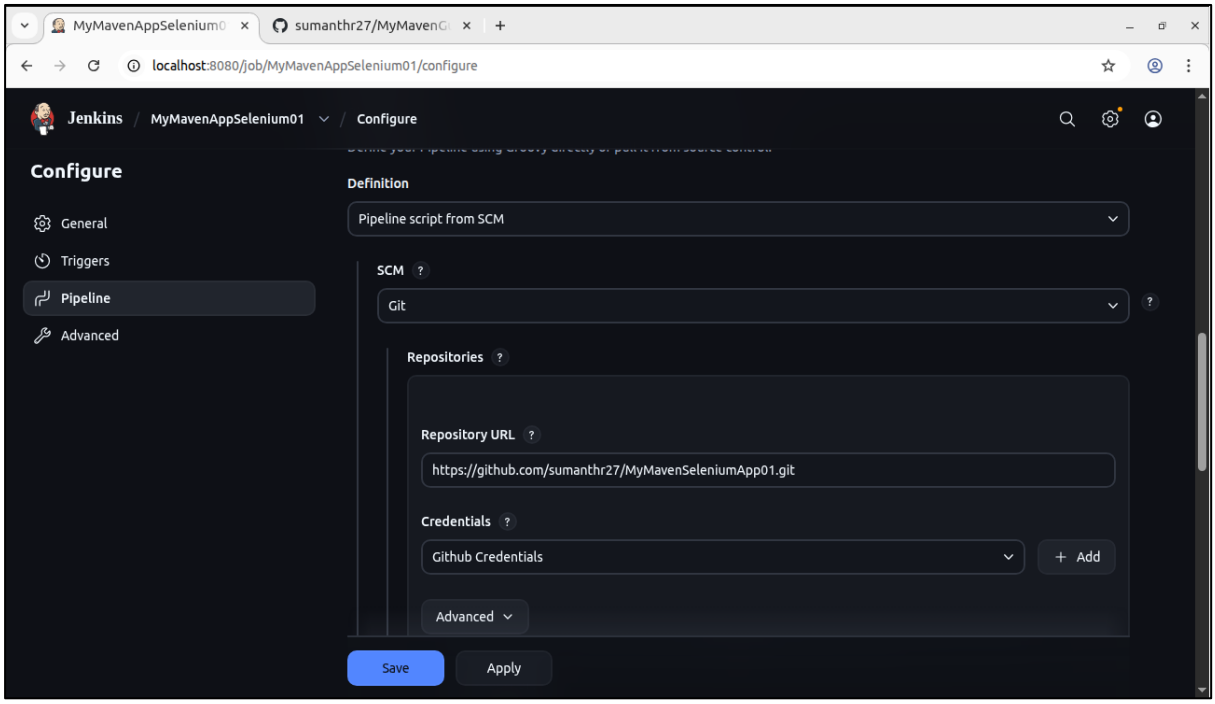


Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenAppSelenium01” and Item type as Pipeline and

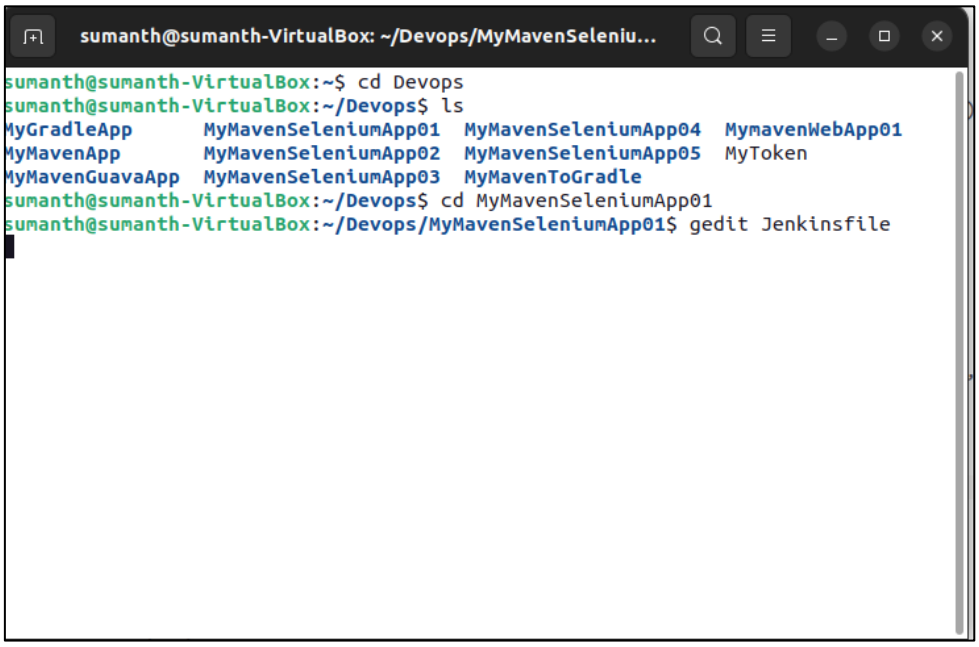
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium01 and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

Jenkinsfile
~/Devops/MyMavenSeleniumApp01

```
pipeline {
    agent any   // Run on any available agent

    tools {
        maven 'Maven'   // Must match name configured in Jenkins (Global Tool Config)
    }

    stages {

        stage('Checkout') {
            steps {
                // Recommended if repo is configured in Jenkins
                checkout scm

                // OR use this if not configured:
                // git url: 'https://github.com/sumanthr27/MyMavenSeleniumApp01.git', branch: 'master'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }

    post {
        always {
            echo 'Pipeline execution completed'
        }
        success {
```

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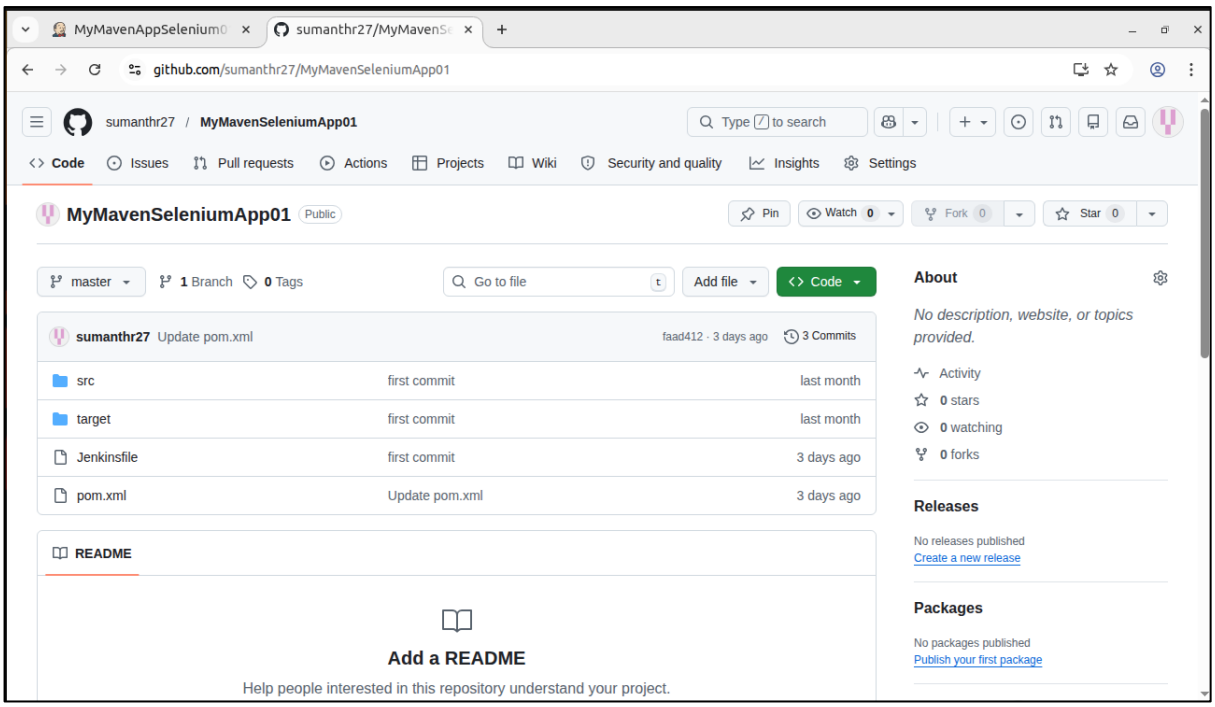
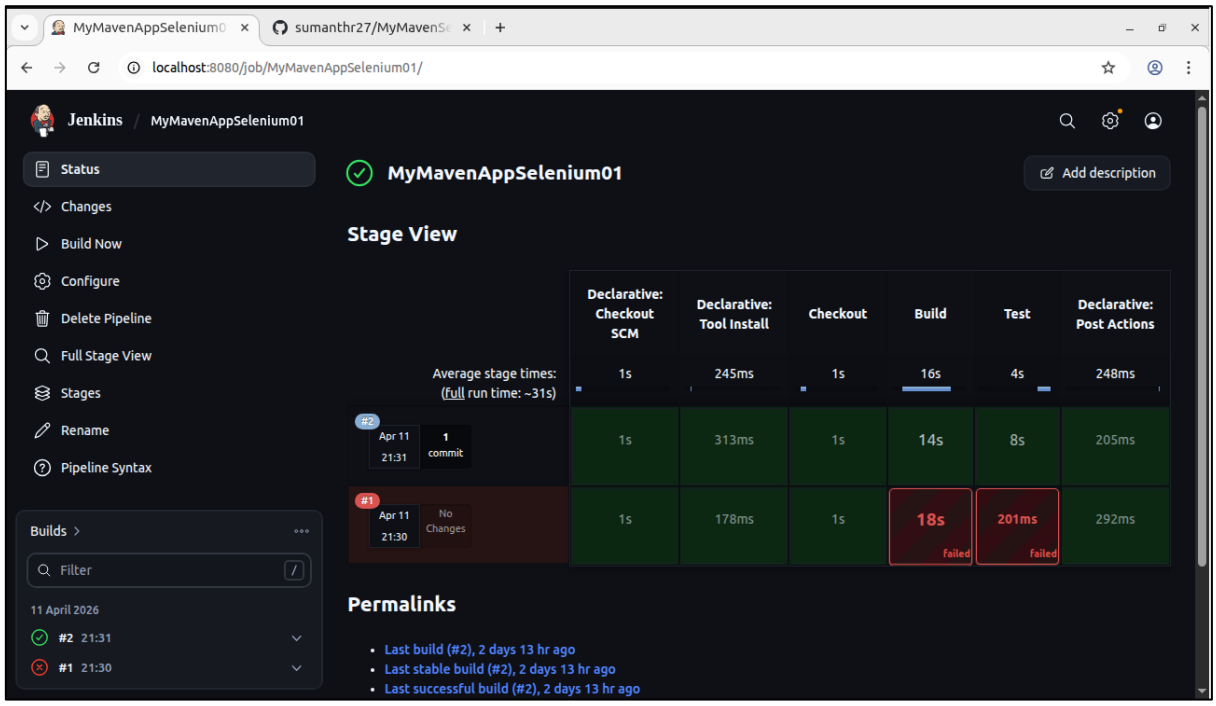
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.